Lei Zhao

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Summer 2027 internship availability: May 17–August 6, 2027 (12 weeks, full-time)

ML System | **Distributed GNN** | **Graph Recommendation System** | **CXL Memory Pool**

EDUCATION

08/24 – Present **Texas State University** *(Expected graduation: May 2028)*
Ph.D. Candidate in Computer Science (Advisor: Dr. Chul-Ho Lee)

09/21 – 06/24 **China University of Geosciences**
M.Eng. in Electronic and Information (Advisor: Longsheng Wei)

09/08 – 06/12 **Hubei University of Technology**
B.Eng. in Electrical Engineering and Automation

WORK EXPERIENCE

05/26 – 08/26 **SK hynix America** – *GNN Advanced Memory System Research Intern*

- **Developed Linux kernel patch** to expose CXL memory as Device DAX on an AMD server without Special Purpose Memory (SPM) support.
- **Developed CXL shared-memory library** with C/C++ APIs and PyTorch bindings, implementing software-managed cache coherence.
- **Offloaded graph and feature data to a CXL memory pool**, substantially reducing GNN training time versus the cuGraph baseline.
- **Investigated CXL+RDMA multipath data transfer** to mitigate workload skew.
- **Explored CXL memory pooling** for PyTorch DDP gradient synchronization using a custom communication hook.
- **Implemented and evaluated experimental CXL-backed NCCL collectives**.
- **Benchmarked NVSHMEM against CXL shared memory**.
- **Evaluated NVSHMEM's UCX transport** for cross-host GPU communication, including GDRCopy integration and memory-registration diagnostics.
- **Explored CXL-backed KV cache offloading** for SGLang prefill–decode disaggregation.
- **Investigated read amplification caused by sparse graph data access**.
- **Set up on-premises LLM coding agent service**.
- **Developed team tools and resolved server issues**, addressing APT update failures, VSCodium remote connection issues, PSU and BMC problems, and incorrect DIMM population order. [Remote Local Network Share]

06/16 – 05/20 **Wuhan Stack Tech Co., Ltd.** – *Founder and Web Developer*

- Developed websites, Chrome extensions, and WeChat mini programs.

07/12 – 05/16 **Ericsson** – *Integration Engineer*

- Performed system integration and network optimization.
- Analyzed signals and verified features against 3GPP specifications.

09/10 – 06/11 **HGDonline.net Coding Club** – *Web Developer*

- Contributed to HBUT BBS redesign and developed student grade portal.

SELECTED PROJECTS

Ongoing **RISCuda: GPU-Resident LLM Tool Execution** [Github]

- Enabled **on-GPU tool calling without CPU invocation or dispatch per tool call**: Qwen generation, tool-call detection, C calculator execution via a CUDA RISC-V interpreter, and result feedback run on the GPU within a **single host CUDA Graph launch**.
- Initial C-calculator tests on RTX A6000 achieved **1.71× speedup over Qwenputer** across 192 jointly correct, normally completed requests; both implementations answered **195**/**200** arithmetic questions correctly.

08/25 – 04/26 **FROST: A Unified Multi-GPU Training Framework for Temporal Graph Neural Networks** [Paper] [Github]
Accepted to ACM SIGMOD 2027.

- Developed a **multi-GPU training pipeline** using multiple threads, processes, and CUDA streams, supporting 8 TGNN models across memory-based and memoryless architectures and achieving up to **12.61× speedup over TGL**.
- **Designed Planner-Syncer system** to analyze cross-batch node-state dependencies and route updated states through same-GPU reuse, GPUDirect transfer, or asynchronous host-memory write-back.
- **Identified an idle inter-GPU communication window** between node-state updates and the first gradient sync, and implemented custom communication hooks to overlap state transfers with remaining forward and early backward computation.
- **Developed shared-memory ring buffers** for zero-copy host-to-worker graph access, scattering only lightweight metadata to avoid bulk graph serialization.
- **Implemented topology-aware NUMA placement**, binding GPU workers and per-GPU shared-memory buffers to GPU-local NUMA nodes before first touch.
- FROST provides a system abstraction for managing mutable per-entity states across GPUs and host memory, with potential applications to other stateful workloads such as recommendation systems.

05/26 – Present **Accelerating cuGraph GNN Training via CXL Memory Pool Data Offloading**
Collaborative Project with NVIDIA cuGraph team

- Reduced per-batch sampling and feature-gathering time compared with the RDMA-based cuGraph baseline.
- **Profiled cuGraph training** to measure per-stage execution time, identify performance bottlenecks, and count sync barriers.
- **Implemented CXLGraph, a data loader, and a neighbor sampler** to offload graph data to a CXL memory pool and eliminate RDMA sync barriers in cuGraph's distributed graph sampling.
- **Replaced NVIDIA's distributed contiguous memory with CXL-backed memory** to offload feature data to a CXL memory pool and eliminate RDMA sync barriers in cuGraph's remote feature gathering.
- **Implemented versioned tensors** to maintain consistency during mutable embedding updates.
- Developed CI test cases, ran CI tests, and contributed to cuDF and cuGraph GitHub repositories (cuDF PR #23195; cuGraph PR #5568).

11/22 – 04/23 **Building Ultra-Low-Cost GPU Computing Servers**

- Built multiple GPU servers, including one with **15 NVIDIA P102-100 mining GPUs** and PCIe splitters.
- Benchmarked GPUs with ai-benchmark and FurMark.
- Configured Xubuntu, networking, and NFS; automated environment deployment with Dockerfiles; and profiled and tuned system performance using monitoring tools, bcache, and swap.

SKILLS

- **Programming:** Python, C/C++, CUDA, OpenMP, Shell.
- **Distributed Training:** PyTorch Distributed, DDP, NCCL, CUDA streams, pipeline optimization, PyTorch profiling, Nsight profiling.
- **Systems:** Linux, CXL/Device DAX, NUMA, shared memory, Docker, Git.

PUBLICATIONS AND MANUSCRIPTS

(🏛 marks pseudonymous titles)

- **Lei Zhao**, Joseph Nke, Jongryool Kim, and Chul-Ho Lee, "**FROST: A Unified Multi-GPU Training Framework for Temporal Graph Neural Networks**," *Accepted to ACM SIGMOD 2027.*
- Author A, **Lei Zhao**, Author C, Author D, Author E, Author F, and Author G, **Accelerating SpGEMM with Near-HBM Processing** 🏛 (*Under review*)
- Author A, **Lei Zhao**, and Author C, **A CUDA Graph Analytics Primitive** 🏛 (*Under review*)
- Author A, **Lei Zhao**, and Author C, **A CUDA Graph Data Structure** 🏛 (*Under review*)
- **Lei Zhao**, Longsheng Wei, Zhi Ma, and Zhiheng Liu, "**A Pavement Crack Segmentation Method Based on Deformable Convolution and Enhanced Perceive Network**," *Multimedia Tools and Applications*, 84(19):20995–21016 (2025).
- Zhi Ma, **Lei Zhao**, and Longsheng Wei, "**Single Human Parsing Based on Visual Attention and Feature Enhancement**," *Journal of Advanced Computational Intelligence and Intelligent Informatics*, 27(4):561–566.
- Longsheng Wei, **Lei Zhao**, and Jian Peng, "**Reduced Reference Quality Assessment for Image Retargeting by Earth Mover's Distance**," *Applied Sciences*, 11(20):9776.

AWARDS

- **At Texas State University:**
  - Graduate Research Excellence Award (2025–2026)
- **At China University of Geosciences (CUG):**
  - First Class Scholarship (Awarded twice), Second Class Scholarship (Awarded once)
  - Third Class Prize in China Graduate Electronics Design Contest
- **At Hubei University of Technology (HBUT):**
  - Second Class Scholarship (Awarded once), Third Class Scholarship (Awarded twice)
  - Outstanding Student

PATENTS

- A Human Body Posture Estimation Method and System Based on Shape Similarity. CN202210752411.0

- A Reference-Free Image Quality Evaluation Method Based on Graph Convolution and Multi-scale Features. CN202310801526.9
- A Twin Network Single Target Tracking Method and Device. CN202310897743.2
- Real-time Semantic Segmentation Method, Device, Electronic Equipment, and Storage Medium for Night Images. CN202310856554.0